Anti-Occlusion Infrared Aerial Target Recognition With Multisemantic Graph Skeleton Model

Xi Yang, Shaoyi Li, Shijie Sun, and Jie Yan

***Abstract*— In photoelectric countermeasure systems, the infrared imaging of missiles is critical for automatic recognition and tracking technology of aerial targets. However, complex and newly emerging infrared interference signals severely hinder the recognition performance and lock the target ability of infrared thermal imaging systems. Although considerable progress has been achieved in the development of machine vision systems for missile detection, their performance and robustness should be improved. The brain can detect learned objects in various nonideal situations (partial occlusion and various perspectives). A novel graph network learning framework was developed for object recognition. This brain-inspired anti-interference recognition model can be used for detecting aerial targets composed of various spatial relationships. A spatially correlated skeletal graph model was used to represent the prototype using the graph convolutional network. Furthermore, a novel anti-occlusion framework based on a multisemantic skeleton graph model was proposed to overcome the discontinuity of target features caused by occlusion. In this method, the location of occluded key points was inferred by learning high-order relationships and node topology information. In this study, local image features were considered as graph nodes and a high-order relationship learning module was proposed to transfer relational information between nodes. In this module, the degree of connection between target keypoints was learned to automatically suppress the delivery of meaningless features. Second, a high-order topology learning module that simultaneously learns topological information and embeds local features was proposed to directly predict node similarity scores. Finally, extensive experiments were conducted on the constructed aerial target flight infrared dataset to validate the effectiveness of the proposed model.**

***Index Terms*— Aerial target, anti-occlusion, graph convolutional network (GCN), infrared seeker, multisemantic, skeleton model.**

I. Introduction

AERIAL target automatic recognition and tracking are critical technologies of infrared-guided missile seekers and are crucial for achieving accurate aerial attack capability. However, as shown in Fig. 1, texture loss, variation in the target scale and attitude, interference in background clutter, and infrared decoy interference severely affect the effectiveness of infrared aerial target recognition. Therefore, realizing accurate and rapid recognition of infrared aerial targets is challenging. The infrared characteristic of aerial targets in complex aerial combat environments results in an inherent lack of color information. Because of the interference of natural clutter, infrared target texture and contour feature exhibit low signal-to-noise ratio and saliency. Furthermore, target feature extraction is disturbed by the background clutter. Therefore, selecting a stable and effective target characterization method is critical in infrared target feature extraction. Studies have revealed that in the human visual system, the shape information of objects plays a crucial role in object recognition [1], [2], [3]. In this study, a novel skeleton model was introduced to represent the target shape structure. The model provides not only a compact description of the overall shape structure of the object but also a quantitative measure for simultaneously comparing visual similarities between shapes.

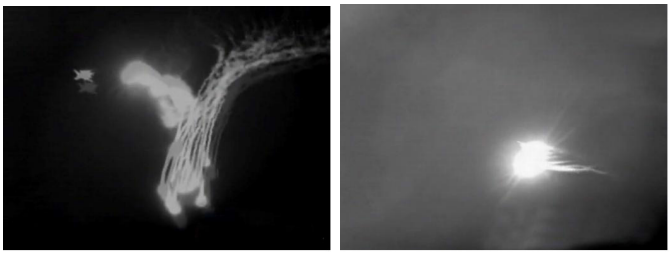


Fig. 1. Spatial relative position relationship between the target and infrared interference.

The artificial infrared decoys dropped by the aerial infrared target cause considerable occlusion on the target, which severely affects recognition and tracking of infrared seekers. In the human visual system, the shape and position of the whole object are predicted using the partial structure information of known objects. The global phase structure is then restored using local shape information. This technique is the closing property of visual perception. Inspired by this characteristic, the target skeleton model was constructed in this study to estimate the structure and position of the skeleton under partial occlusion and realize target anti-interference recognition.

The ability to determine the similarities between shapes is crucial for aerial recognition. Shape skeletons may play a crucial role in context recognition because they provide compact descriptions of shapes and formal methods for calculating the shape similarity [4]. However, limited bio-inspired recognition

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models have incorporated skeletal structure descriptions in their implementations.

In humans, keypoint localization is a two-stage process [5], [6], [7], [8]. The target key location is first located based on context information. This step is the proposal process. After approximate localization, the detailed structure of the keypoint is observed to determine the exact position of the keypoint. This process is called the refinement process. In this study, inspired by the shape perception characteristics of the human visual system, a visual skeleton information processing model was developed by incorporating the two independent subnetworks of human visual systems for feature extraction and anti-interference recognition of targets. In the proposed method, first, the skeleton structure representation of the object is constructed and stored along the hierarchical structure of the model. Next, the visible part of the occluded object is processed along the model for recognition. The occlusion part then encodes the location information according to the recognition model and adjusts the anti-occlusion network to reconstruct the incomplete expression of the occlusion part. The top-down cues provided by direct connections between the lower and higher levels of the ventral stream are simulated to achieve this recognition.

Deep convolutional neural networks (DCNNs) have attracted considerable research attention in computational vision and neuroscience. These networks are consistent with the fast, feedforward visual dynamics of the brain and based on simple and straightforward neuronal assumptions. DCNNs consist of multiple layers of features, which are arranged in a feedforward hierarchy. DCNNs are trained differently to optimize the recognition of target classes from labeled data. Based on the role of DCNNs in predicting the neural responses of the human visual systems [9], [10], [11], [12], a target keypoint inference model was constructed from the feedforward network instead of the top-down object recognition or object classification such as the conventional DCNNs. We constructed the bottom-up backbone network in which only a bottom-up traversal of the image is required. This model does not require considerable iterative processing.

Second, in addition to simple localization, a comprehensive approach is required to uncover occluded nodes. A key insight is that invisible joints are closely related to the contextual understanding of images and structural understanding of the attitude of aerial targets. Humans can easily infer the location of invisible joints from cues about the maneuver type and image context. To reduce the meaningless features caused by the occlusion of keypoint information of the target, we studied the cues required for the inference of invisible joints and proposed a novel framework of joint high-order relationship and target skeleton topology information for occluded target recognition.

The proposed framework operates in two stages, namely, initial keypoint estimation and graph convolutional network (GCN)-based pose correction. In stage 1 of the model framework, the first-order semantic module, that is, the spatial correlation graph (SCG) module, is used to generate the initial pose with the generated heatmap. The subsequent correction stage includes a higher order relationship learning (HRL) module and a higher order topology learning (HTL) module. The image-guided progressive GCN module is used to adjust and refine the initial pose obtained from the heatmaps.

1) In SCG, the backbone network is used to learn interest points of the target. The semantic features of the corresponding keypoints are then extracted.

2) In HRL, the learned semantic features of images are considered the nodes of graphs, and an adaptive direction graph convolution layer was proposed to learn and transfer the information of edge features and automatically determine the direction and degree of each edge. Thus, information transmission of semantic features is promoted, and information transmission of meaningless and noise features is suppressed. Finally, the learned nodes contain semantic information and related information.

3) In HTL, a graph embedding prediction (GEP) layer is used to fuse the structural clues of image features and target posture, learn the corresponding relationship of nodes on the coincidental graph, capture the interaction relationship by connecting corresponding nodes, and deduce the occluded nodes of the target. Finally, to easily achieve strict one-to-one alignment, their similarity is predicted by mapping the coincidence graph and monitored to detect loss of validation.

The main contributions of this study are as follows.

1) Based on the sensitivity of the human vision system to skeleton structures and the completeness of description of global and local location attributes of the target, a novel framework was proposed for target anti-interference recognition to combine high-level semantic relationships and topological information of the skeleton graph.
2) A bottom-up feature extraction backbone network was constructed based on the structural features of the skeleton graph, and the points of interest and their features were identified by extracting targets. A spatial correlation module SCG was proposed to constrain the relationship between nodes and realize the skeleton graph model of the target. This model can be adapted to detect the variation in target maneuvering, scale variation, and achieve feature representation of local structures.
3) Based on the perceptual closure characteristics of human visual systems, two high-order semantic learning modules, namely, HRL and HTL, were proposed to dynamically learn the connection form of the target skeleton graph, promote the message transmission in the semantic region, and inhibit information transmission in the occlusion region. The visible part of the target was used to encode position information according to the recognition model and adjust the anti-occlusion network to reconstruct the complete representation of the occlusion part.

II. Related Work

A. Shape Description Based on Skeleton Structure

Numerous studies have revealed that the primate visual systems extract skeletal structures during shape perception [13], [14]. Behavioral studies have indicated that

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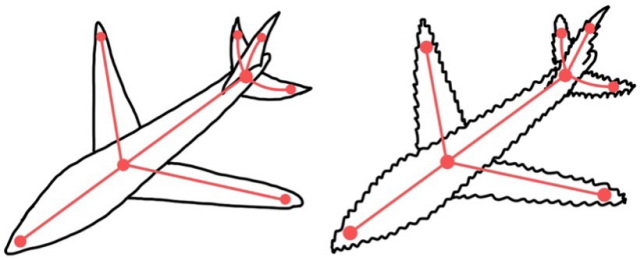


Fig. 2. Schematic of the aerial target skeleton.

the visual system exhibits not only considerable sensitivity to the bone structure of objects but also the perception and comparison of the bones of objects [15], [16]. Human representations of the skeleton structure are robust to view changes, contour disturbances, and deformation caused by bending or stretching. Thus, global shape attributes, such as the skeleton structure, are more dependent than local contour information, as displayed in Fig. 2. Behavioral studies [17] have revealed that the skeletal structures of various shapes extracted by human participants can remain stable in case of target edge damage caused by interference.

Numerous studies have revealed that skeletal structures can be represented in three dimensions (3-D) within an object-centered reference frame. Humans can accurately distinguish 3-D objects by the differences in skeleton structure rather than by the differences in parts [18]. Furthermore, neural recordings (i.e., functional magnetic resonance imaging and electrophysiological experiments) of humans and monkeys have revealed that the areas of higher visual cortex, including those known to support object recognition, are highly sensitive to the skeletal features of 3-D objects [15], [16]. The skeleton feature sensitivity of these regions was decoded by the variations in orientation and local shape attributions, which indicated that the 3-D object-centric representation was robust to the changes in the target.

B. Pose Estimation Model Based on Heatmaps

The attitude estimation models can be categorized into bottom-up and top-down models. In the bottom-up approach, the keypoints were first detected and subsequently assigned to matching targets. Numerous joint grouping strategies have been proposed [19], [20], [21], [22]. In the top-down approach, first, all the targets in the scene are detected. Subsequently, critical point estimation for each target is then performed. Most existing top-down methods [23], [24], [25] have focused on devising a efficient target detector to obtain superior results. However, none of these strategies can adequately reduce errors, especially in the case of severe occlusion in which a candidate box captures the keypoints of multiple targets. Most mainstream methods are based on the heatmaps and limited to estimating invisible joints and lacking visual information. Therefore, a novel multisemantic graph skeleton model (MSGSM) was proposed to estimate the invisible target keypoints by inference rather than through localization to achieve robust target recognition in aerial scenes with complex background and artificial interference.

C. Keypoint Model Based on the Graph Convolution Network

Modeling keypoints as graph problems can be traced to the first studies on scene understanding [26], [27]. Studies focused on graph representation have used belief propagation to solve pose estimation [28], [29]. In studies [30], [31], [32], classical graph modeling has been extended to the deep learning paradigm to achieve excellent modeling capabilities for unstructured data. Some advanced techniques, such as motion recognition [33], motion prediction [34], and 3-D posture regression [35], [36], have focused on constructing graphic networks to solve problems related to target posture. Compared with other approaches, GCNs exhibit a considerable advantage when addressing target pose modeling problems. These techniques can capture the dependencies between joints effectively. In previous studies [33], [34], spatial and temporal relationships have been used as edges in graphs. However, the GCN framework is yet to be introduced into critical point detection problems such as infrared targets. In contrast, the graph network is specifically designed for keypoint detection of infrared targets and contains a progressive learning strategy guided by image features.

III. Skeleton Graph Model of the Aerial Target

Visual neuroscience studies have revealed that the shape of an object is a critical factor affecting object recognition [4]. In the absence of other visual information (textures, contour, and shadows), humans can identify objects by their shapes. Shape skeleton is a geometric model based on the shape axis. These models describe shapes that pass between two or more points equidistant along a boundary along a symmetric axis. Specifically, a skeleton describes the shape and structure of objects by specifying the outline and spatial configuration of its parts. The skeleton model is an effective method for describing the global shape structure of objects.

A. Keypoint Extraction

In stage 1, first-order semantic features of keypoint regions, which are inspired by two cues, are extracted. Part-based features have been proven effective for occluded object recognition. Second, an accurate alignment of local features is necessary in partial occlusion of objects [37], [38]. Inspired by the prediction of keypoints [39], [40], local features of various keypoints are extracted based on the convolutional neural network (CNN) model. Although keypoint predictions have achieved high accuracy, their performance remains unsatisfactory in occlusions/partial images [41].

In the bottom-up target detection process [42], [43], [44], points of interest are extracted for each frame. This phenomenon is consistent with the human cognition and recognition process of visual targets. Each aerial object detection in each frame is first cropped and subsequently scaled to a fixed size. In the interest point detection algorithm, the scaled image is considered as the input, and the output consists of two parts, namely, the interest point heatmap and the pixelwise interest point regression result. The interest point heatmap $\mathbf{H}$ consists of K maps $\{\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_K\}$, and the number K of interest points in this dataset is set to 9. The heat value at each location

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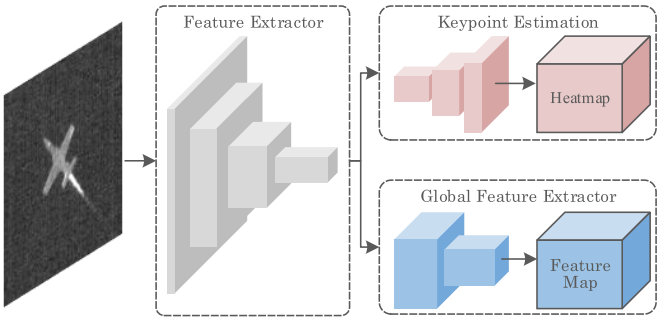


Fig. 3. Stage 1: keypoint feature extraction.

in the keypoint heatmap represents the probability of the point of interest. As shown in Fig. 3, the pixelwise interest point regression results consist of the following two parts. Center heatmap C represents the degree to which each location is the center of a pose. The offset map O represents the offset of the interest point to the center of the object. The value of the position (i, j) of the kth heatmap $\mathbf{H}_{ijk}$ is the probability value of the position of the kth interest point. The position of the kth point of interest can be calculated as follows:

$$p_k^* = \arg\max_{(i,j)} \mathbf{H}_{ijk} \tag{1}$$

where p_k^* is the location of the target keypoint in the heatmap, which can be converted to a location in the image based on the center and scale information of the cropped image. The training loss of the keypoint detection model is calculated according to the heatmap. After cropping, the target image is scaled to a fixed size, and the corresponding annotation data are appropriately transformed to the coordinates in the heatmap. Let p_k^* be the standard position of the kth node in the heatmap. Standard heatmaps were generated according to a 2-D Gaussian distribution model. In all the experiments, σ was set to 3. The keypoint detection model was trained by minimizing the subsequent loss. The loss function in stage 1 of the algorithm training process is expressed as follows, where H and W represent the height and width of the heatmap, respectively, and K is the number of keypoints:

$$L^{s1} = \sum_{i}^{H} \sum_{j}^{W} \sum_{k}^{K} \left\| \mathbf{H}_{ijk}^{\mathrm{dt}} - \mathbf{H}_{ijk}^{\mathrm{gt}} \right\|_2^2. \tag{2}$$

B. Representation Method of the Skeleton Graph Model

The human visual system responds selectively according to the part of interest of an object [45]. To encode various objects, these target prototypes are decomposed into many regions of interest. The brain decomposes an object into visually distinct and spatially nonoverlapping regions called the regions points of interest. Each point of interest is considered to be a building block in modeling. Therefore, for modeling purposes, interest points are multiscale representations that represent the various appearances of objects in various situations. Interest points are used for constructing a graph model of the target skeleton structure.

In this study, motivated by work in behavioral and neuroimaging studies, the sensitivity of the primate visual systems to skeletal structure and its increasing relevance were investigated. The representation of 3-D skeletal structures in 2-D images was studied. Object perception appears to be the best method to describe skeleton structures in terms of 3-D object-centric shape representations [14]. Therefore, a novel target association model framework was proposed based on a graph model in which the skeleton structure model was used for realizing target shape representation. The model can be used to accurately and robustly measure the similarity between the target and noise and interference.

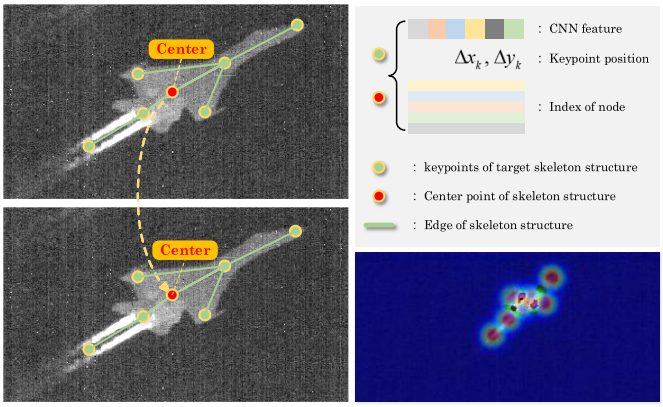


Fig. 4. Construction of the node features of the skeleton graph model.

Each node of the constructed skeleton graph model is represented by a heatmap of the associated feature space [23], [46]. As mentioned in Section III-A, the skeleton graph model consists of K interest points. Therefore, an intuitive graph $G(V, E)$, where the vertex set V of the graph corresponds to Section III-A, was constructed and $V = (V_i | i = 1, 2, \ldots, K)$ to be K nodes. The relationship in which a physical connection exists between two interest points in the skeleton graph is encoded as an edge set $E_{ij} = \{V_i, V_j\}$. The adjacency matrix $\mathbf{A} = \{a_{ij}\}$ of G is expressed as the connection relationship between nodes, and the expression of a_{ij} is as follows:

$$a_{ij} = \begin{cases} 1, & V_i \text{ and } V_j \text{ are neighbors or } i = j \\ 0, & \text{others.} \end{cases} \tag{3}$$

After determining the expression form of the skeleton graph model, three features were combined at each node to construct the input node features, including keypoint feature v_k of the backbone network of a single-frame image, code c_k of its interest point type in the lookup table, and second confidence score p_k for dimension location, where k represents the index type of the kth point of interest for a given target, and Fig. 4 details how the node features are constructed. Each interest point captures specific target local features and is associated with a feature vector v_k, which is a set of node feature maps derived from the local regions of the underlying image to provide additional high-level pose information. The feature is obtained by extracting the heatmap $\mathbf{H}_k$ of each keypoint and correspondingly multiplying it with global feature $\mathbf{J}$ to obtain the target pose guidance feature $\mathbf{F}_k$, and each feature corresponds to a keypoint of the target. The keypoint region is quantized to obtain the feature vector $v_k \in \mathbb{R}^{D_v}$, where $D_v = 1024$. The multilevel pose information can be combined as an attention mechanism that the network adaptively focuses

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on the target area. Multilayer perceptron (MLP) can be used to transform all the joint features to have the same dimension. These joint features are merged into an average pool, that is, the final feature of the kth joint is calculated as follows:

$$\mathbf{V}_k^l = \text{Pooling}\left(\mathbf{MLP}_{\text{vis}}(v_k), \mathbf{MLP}_{\text{pos}}(p_k), \mathbf{MLP}_{\text{ind}}(c_k)\right) \quad (4)$$

where various MLP encoders do not share parameters with each other.

C. SCG Module

The visual system not only perceives the points of interest but also exhibits stable spatial relationships with each other, which remains unchanged under various perspectives and are identified and labeled in the visual cortex according to their stable spatial locations [47]. Objects are identified as a specific class if and only if the structural information is sufficiently close to that of the prototype. In object detection or recognition, the system detects and indexes the target parts and the structural relationship between them. Subsequently, these objects are compared with the prototype to determine the target category and their location. Therefore, an SCG module was proposed to simulate the human visual system: the nodes of the graph model are the target keypoints, and the edges impose relative distance and scale/size constraints. Each pair of interest points V_i and V_j is represented by the parameter e_{ij} of the edge as a constraint of relative variation, which maintains necessary associations required for target integrity.

Because the set of nodes V and the direct interactions E between them constitute a spatial association graph network, the complexity of the skeleton structure graph model corresponds to its sparsity, which has physical meaning in the skeleton model constructed in this study: for most physical objects, the position of parts and the resulting views are not statistically fully connected. A sparse set of interactions is sufficient for constraining target deformation. A sparsity constraint is introduced in the learning process, and an optimization problem that can be solved efficiently is obtained.

An SCG module is a set of variable nodes with spatial structure constraints, which is described by an undirected graph. In particular, each graph variable node is independent of other random variables in the network given its neighbors. Simultaneously, the target skeleton model is a spatial association graph network of the target local nodes. Fig. 5 describes the overall framework for constructing a spatial association graph model. Next, the SCG module of the target is constructed using a GCN.

GCNs are designed to efficiently learn layer-by-layer propagation operations that can be directly applied to graph data types. According to GCN definition [48], given an undirected graph $G(V, E)$, the graph includes K nodes V and edges between connected nodes. The node connection relationship represented by the edge is expressed as the adjacency matrix $\mathbf{A} \in \mathbb{R}^{K\times K}$, and the degree matrix $\mathbf{\Lambda}_{ii} = \sum_j \mathbf{A}_{ij}$ of the graph. The linear transformation of the graph convolution is expressed as the product of the graph signal $\mathbf{X} \in \mathbb{R}^{D\times K}$ and the filter $\mathbf{W} \in \mathbb{R}^{D\times C}$, where the column vector $\mathbf{X}_{i,} \in \mathbb{R}^D$ is

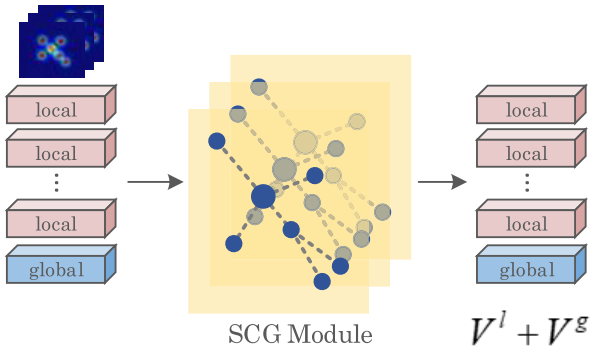


Fig. 5. SCG model.

the feature representation of the ith node as follows:

$$\mathbf{V}^{\text{out}} = \hat{\mathbf{\Lambda}}^{-\frac{1}{2}}\hat{\mathbf{A}}\hat{\mathbf{\Lambda}}^{-\frac{1}{2}}\mathbf{X}^{\top}\mathbf{W} \quad (5)$$

where $\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, where $\mathbf{I}$ is the identity matrix, and $\hat{\mathbf{\Lambda}}_{ii} = \sum_j \hat{\mathbf{A}}_{ij}$. The size of network output matrix V is $C \times K$. The GCN can be formed by stacking multiple graph convolution layers as expressed in (5), and each layer is followed by a nonlinear operation [such as a rectified linear unit (ReLU) layer]. The spatial relationship and constraints between nodes are captured according to the obtained local features and attributes of the target, and the node features are input into the graph convolution module to represent the relationship features between skeleton graph nodes as follows:

$$\mathbf{V}^S = \phi_g\left(\mathbf{V}^l\right) = g_1\left(G \otimes \mathbf{V}^l\right) + g_2\left(\mathbf{V}^l\right) \quad (6)$$

where $\mathbf{V}^l$ represents the local node feature, $\mathbf{V}^S$ represents the output as the node relationship feature, and g represents the fully connected layer. Because the edges in this spatially correlated graph model are connections between nodes within the same skeleton, the connection form of nodes within the same frame enables the GCN to capture the relative motion and spatial structure of target interest points. To encode variations, edges are represented as connections between nodes with stiffness and length variations. Because not all the interest points are directly connected to each other in a physical object, the model considers sparsity: the model only introduces spatial constraints between key pairs of interest points required to maintain the integrity of the entire object, that is, the presence of physical constraints is introduced between connected pairs of interest points, and the adjacency matrix of the corresponding skeleton graph exhibits sparsity.

IV. MSGSM for Anti-Interference Recognition

Each keypoint heatmap is obtained according to the backbone network in stage 1 and subsequently directly decoded into the interest point coordinates using the heatmap. However, stage 1 produced unsatisfactory results on invisible joints because locating the occluded target parts from the heatmap is highly challenging. This phenomenon can be attributed to the fact that the local key parts of the target are occluded and do not exhibit visual information. Because the first-order node features extracted in stage 1 do not consider two characteristics of aerial targets: 1) various features and processes are used for accurate localization and 2) the relationship between keypoints

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should be considered. To remedy this deficiency, a novel framework was proposed to infer invisible joints from the image background and propose structural cues.

In human vision, keypoint localization is a two-step process [5], [7], [8]. For example, for the green joint in Fig. 4, including the two wings displayed in the yellow circles, rough localization based on contextual information determines whether wing keypoints are present in the nearby area. This step is an approximate localization process. In human vision, the perceptual closure feature is used to perform object location estimation, and even in occluded situations, the exact location of occluded parts can be identified from the action type and image context. After rough localization, the cues required for the inference of occluded parts were studied. The detailed structural relationship of the target was established to learn the high-order topological relationship of the target skeleton structure so that the occluded area can be predicted in the case of severe occlusion. This process is the refinement process.

In the object detection method, localization and refinement are performed based on two feature maps of two independent subnetworks (Fig. 7). The keypoint rough localization and refinement process should also be based on various feature maps. Therefore, two subnetworks were used to obtain feature maps for localization and refinement. Furthermore, two-stage methods are typically used in object detection for effectiveness and performance. Therefore, a two-stage design is applied to the keypoint prediction task, in which stage 2 is focused on the refinement process to improve prediction and localization accuracy. This section describes stage 2. According to the features corresponding to the selected guide points in stage 1, the coordinates are refined to achieve accurate regression of target key parts.

A. High-Order Relationship Learning Module

Although the first-order semantic information of various keypoint regions is known, because of the incomplete target structure, the recognition of anti-occlusion targets is difficult. Therefore, higher order semantic features should be investigated and GCN methods [49] should be used to model higher order relational information. A novel HRL was proposed. In this module, the first-order semantic features of various regions of interest are considered as nodes. The structural relationship of key parts of the target can be incorporated into the local representation to predict occluded regions. When passing messages between nodes, not only first-order semantic information (node features) but also higher order relational information (edge features) can be considered simultaneously. However, occlusion region prediction remains a challenge. The features of occluded regions are often meaningless features. When passing disturbing features in a graph network, noise is added and a side effect occurs on occlusion region prediction. At this stage, the node interaction information is crucial for predicting the position. Therefore, a novel adaptive graph convolution (AGC) layer was proposed. In this module, the direction and degree of message passing are directly learned to automatically suppress the passing of meaningless features and improve the passing of semantic features as follows:

$$\mathbf{V}^{\text{out}} = \hat{\mathbf{A}}\mathbf{X}\mathbf{W} \tag{7}$$

where $\mathbf{A}$ is the predefined graph adjacency matrix, $\hat{\mathbf{A}}$ is the normalized $\mathbf{A}$, $\mathbf{X}$ is the input graph node feature, and $\mathbf{W}$ is the parameter. By improving a simple graph convolutional layer, the input features of the adjacency matrix (links of nodes) can be adaptively learned. Assuming that given two local features, the meaningful features are closer to the global features than the nonmeaningful features. Invisible keypoints are then extracted. Because local node features lose more contextual information, local features and global features are combined to strengthen the association of contextual information. Therefore, an AGC layer, whose inputs are global feature $\mathbf{V}^g$ and local feature $\mathbf{V}^l$, and a predefined node adjacency matrix were proposed. The edge weight graph of all the nodes is dynamically updated using the difference between $\mathbf{V}^l$ and $\mathbf{V}^g$, and the result is $\mathbf{A}^{\text{agc}}$. A simple graph convolution is obtained using the product of $\mathbf{V}^l$ and $\mathbf{A}^{\text{agc}}$. To stabilize training, the input local features $\mathbf{V}^l$ are combined to the output of the AGC layer. The AGC layer is expressed as (8), where g_1 and g_2 are two unshared fully connected layers

$$\mathbf{V}^{\text{out}} = \left[g_1\left(\mathbf{A}^{\text{agc}} \otimes \mathbf{V}^l\right) + g_2\left(\mathbf{V}^l\right), \mathbf{V}^g\right]. \tag{8}$$

Finally, the HRL module ϕ_r is implemented as a cascade of AGC modules. Therefore, given an image I, its semantic feature $\mathbf{V}^S = \{v_k^S\}_{k=1}^{K+1}$ can be obtained by (6), and its relational feature map $\mathbf{V}^R = \{v_k^R\}_{k=1}^{K+1}$ can be expressed as follows:

$$\mathbf{V}^R = \phi_r\left(\mathbf{V}^S\right). \tag{9}$$

Classification and triple losses were used to calculate the loss function, as expressed in (10), where α_k is the confidence of the kth interest point of the target, $p_{v_k^r}$ is the probability that the feature v_k^r predicted by the keypoint belongs to its real keypoint category, β is a margin, and $d_{v_k^r}$ is the distance between v_k^r and the ground truth of the same class

$$\begin{aligned} L^{s1} &= \frac{1}{K+1}\sum_{k=1}^{K+1}\alpha_k\left[L_{\text{class}}\left(v_k^r\right) + L_{\text{tri}}\left(v_k^r\right)\right] \\ &= \frac{1}{K+1}\sum_{k=1}^{K+1}\alpha_k\left[-\log p_{v_k^r} + d_{v_k^r} + \beta\right]. \end{aligned} \tag{10}$$

B. High-Order Topology Learning Module

In most studies, the estimation of target keypoints is a simple matching strategy. The same keypoint features are matched directly. However, this first-order alignment strategy localizes each keypoint independently and ignores the relationship between various keypoints. Thus, challenging situations, especially with severe occlusion, cannot be addressed [41]. According to the skeleton graph model constructed in Section I, the target keypoints develop a graph structure based on aerial pose changes. They then exhibit a clear adjacent relationship between them. The keypoint locations can be inferred using the structural information implied by the spatial relationship. For example, in the skeleton graph model of this study, if the guiding point is on the left wing, then the positive

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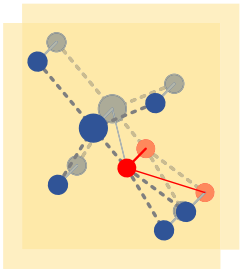


Fig. 6. Coupling connections of the skeletal structure.

samples of the left wing tend to have a higher response to the position of the left wing. Therefore, the keypoint features can be applied to multiple constraints. Because the positions of occluded nodes estimated from the keypoint prediction module differ from their correct positions, estimating their positions directly is challenging. To achieve refined coordinates based on heatmap peaks, coordinate refinement training is performed on node-guided features in stage 2. Therefore, based on the inspiration of the closed feature of human visual perception, automatically completing the target shape in partial occlusion was simulated, and a coarse-to-fine learning mechanism was designed based on the node coordinates. An HTL module HTL was then constructed. In this module, we proposed a novel GEP layer to fuse image features with structural cues of the target pose, perform occlusion node reasoning on the target, and progressively enhance multiple scale image features to stabilize performance. First, based on the relationship feature map of each node position mined by GCN, the network was fused into the module, that is, the pose estimation result is improved by adding the image feature map. Finally, the visual features of the initial nodes and the relation features obtained in Section IV-A are packaged and input into the GEP layer. Specifically, we designed cascaded GEP modules to obtain useful information stored in relational feature maps but lost in the initial pose. Each GEP layer consists of three GCN layers, as displayed in Fig. 6. One GCN layer was used to resize the input features before adding the outputs of the lower two GCN layers. The operation is defined as follows:

$$V^{\text{out}} = \text{GEP}\left(V^R, V^S\right). \tag{11}$$

To capture more node interactions and occlusion joint reasoning, the target skeleton structure graph is extended to a coincidence graph, and the interaction information is captured by connecting the corresponding nodes. The coincidental graph is denoted as $G'(V', E')$, where the number of nodes is 2K, which can be denoted as $V = (V_i | i = 1, 2, \ldots, 2K)$. Here, $=E'$ contains two types of edges, one edge represents the connection relationship of the nodes in the target skeleton, denoted as $E_a = \{V_i, V_j | i \text{ and } j \text{ are connected}\}$, and the other edge is represented as the interactive edge $E_c = \{V_i, V_{i+K}\}$ between the targets. Furthermore, i and $i+K$ connect the same parts of the two targets, as displayed in Fig. 6. The position and type of unknown nodes are inferred using the relational features of the known skeleton structure graph, and the initial keypoints are refined finely.

After the above refinement module, the embedded features are sent to the modules of multiple fully connected layers and normalization layers, and finally two predictions are output as c_k and r_k, which represent classification and regression, respectively. Given the standard keypoint locations, the loss for the two prediction outcomes is defined as follows:

$$L^{\text{cls}} = \exp\left(-\frac{(s_k - t_k)^2}{2\sigma^2}\right) \cdot L_{\text{cls}}(c_k) \tag{12}$$

$$L^{\text{reg}} = L_{\text{reg}}(r_k, t_k - s_k) \tag{13}$$

where L^{cls} and L^{reg} are the soft-max cross-entropy loss and L1 loss, respectively. The total loss for stage 2 can be expressed as follows:

$$L^{s2} = \frac{\sum_k \alpha_k \left(\lambda_2 L_k^{\text{cls}} + \lambda_3 L_k^{\text{reg}}\right)}{\sum_k \alpha_k} \tag{14}$$

where α_k is the target weight of the keypoint k, and λ_2 and λ_3 are the loss weights. In the training phase, the overall objective function of MSGSM is expressed in (15), where L^{s1} is the conventional heatmap regression loss in stage 1, and λ_1 is the loss weight in stage 1. The algorithm framework of this study is trained by minimizing L as follows:

$$L = \lambda_1 L^{s1} + L^{s2}. \tag{15}$$

C. Anti-Occlusion Recognition Framework

The framework of the anti-occlusion recognition network consists of two main stages, namely, visible interest point detection and occluded interest point prediction as displayed in Fig. 7 for the overall architecture of the multisemantic recognition framework. The interest point graph neural network estimates the 2-D image location of occluded interest points by deforming the graph nodes. The modules proposed in this study were combined to accurately predict the location of interest points. In stage 1, a rough localization was obtained using the heatmap regressor, and node relationship features were learned through an SCG module. In the stage 2, guide points with the corresponding relational features were constructed as pose graphs and sent to the refinement module to obtain refinement results. Each stage is described below.

The anti-occlusion recognition network consists of two parts, namely, initial interest point heatmap estimation, high-order graph structure learning, and a high-order graph node refinement module to infer the location of occluded interest points. The proposed method follows the target skeleton extraction method in stage 1 to generate keypoint maps. In bottom-up keypoint estimation methods, the objective of the detector is to estimate K keypoints in an image $\mathbf{I}$ of size $W \times H$. The heatmap-based methods transform the problem into estimating K heatmaps $\{\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_K\}$ of size $W \times H \times K$, where each heatmap $\mathbf{H}_k$ is decoded to the corresponding coordinate p_k^* during the testing part. First, we extract visual features $\mathbf{V}$ from image $\mathbf{I}$ using backbone $\phi_b(\cdot)$ as follows:

$$\mathbf{V} = \phi_b(\mathbf{I}). \tag{16}$$

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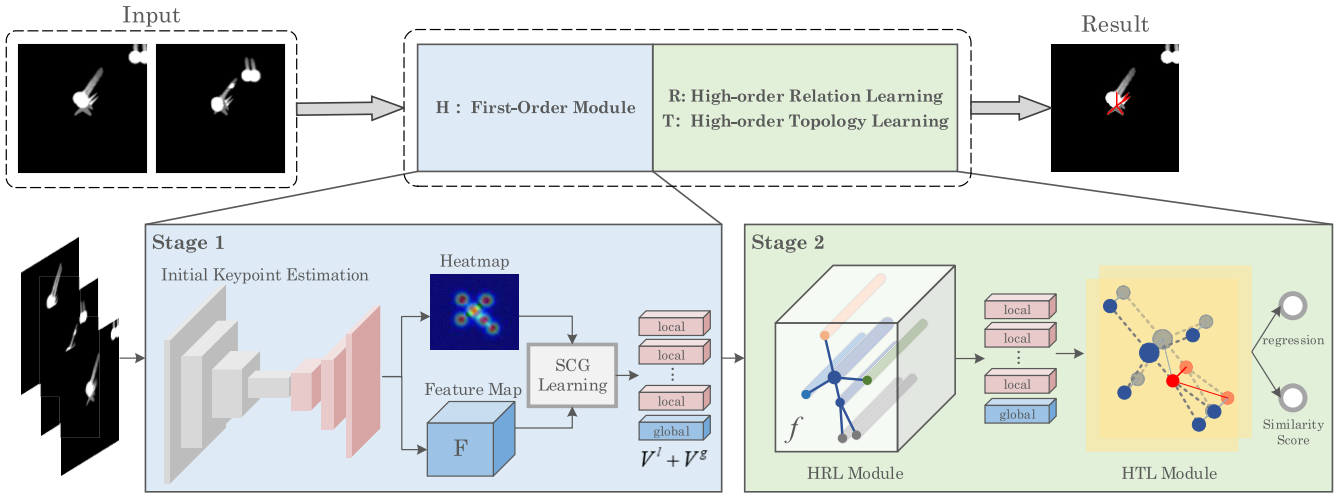


Fig. 7. Overall structure of the two-stage anti-interference recognition network.

A heatmap regression function $\phi_h(\cdot)$ is used to estimate the heatmaps of the keypoint, usually ending with a 1×1 convolutional layer as follows:

$$\{\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_K\} = \phi_h(\mathbf{V}). \tag{17}$$

The anti-interference recognition model is added after heatmap regression. Therefore, this network framework can be applied to any heatmap-based model. The overall structure of the model is displayed in Fig. 7. First, the target skeleton guidance feature is constructed according to (4). Next, the spatial association graph module is used to extract the relational feature graph of the skeleton graph nodes as follows:

$$\mathbf{V}^S = \phi_g\left(\mathbf{V}^l\right). \tag{18}$$

In stage 2, skeleton local feature $\mathbf{V}^l$ and the extracted global feature $\mathbf{V}^g$ are input into the HRL module, and the edge weights are dynamically updated according to the difference of (10) to effectively learn the high-order relationship information between nodes. Finally, according to (12), the mapping relationship between the node relationship feature and the visual feature is mined, the high-order topology learning module HTL is introduced, the reasoning method of the skeleton structure coincidental graph is adopted, and the refined classification result c_k and offset regression result r_k are obtained based on the refined feature g_k to realize refined correction of the initial keypoints. The final refined coordinates of the keypoints are presented in (19), where p_k^* is the position of the target interest point in the initial heatmap

$$p_k = p_k^* + r_k. \tag{19}$$

V. Experiments

Extensive quantitative and qualitative experiments were performed to verify the validity and accuracy of the proposed MSGSM. A comprehensive ablation study was performed to verify the effectiveness of each component.

A. Implementation Details

To verify the effectiveness of the proposed model against infrared decoys, the air combat simulation platform developed by the laboratory based on Microsoft Visual Studio was used to develop an air combat simulation dataset. The specific parameters of the simulated infrared image are as follows: the infrared band is the medium wave (3∼5 μm), the frequency is 100 Hz, and the resolution is 256×256. The air combat confrontation simulation platform is constructed by digital modeling and real-time rendering of infrared scenes, which include the following.

1) Target, infrared interference, and background modeling use Fluent software to complete the flow field calculation and model verification through the field measured data.
2) The infrared real-time scene rendering approach and the output of the simulation dataset use the open scene graph rendering driver and the OpenGL parallel rendering method to generate the infrared scene. Moderate Resolution Tanssios (MODTRAN) software is used to select the atmospheric mode and configure the relevant atmospheric parameters. The target infrared model is imported into the graphic processing unit (GPU) for parallel computation.
3) The confrontation process between the target and the missile includes the design and parameter quantification of the confrontation situation of the simulation system under various conditions and scenarios and includes missile movement model, interference infrared simulation model, and interference movement model.

The generated dataset consists of 36 sequences that cover various challenging situations such as target maneuvering and turning, target attitude rotation, and target maneuvering accompanied by infrared artificial interference. We annotated 73 584 target keypoint coordinates for a total of 8176 targets.

The measured infrared target flight dataset collected by the infrared thermal imagers was used to verify that the proposed model can cope with challenging situations such as environmental interference and target maneuvering in real-world scenarios. The specific parameters of the measured infrared image are as follows: the infrared band includes medium wave (3∼5 μm) and long wave (7.5∼14 μm), a frequency of 100 Hz, with a resolution of 640×512. This result is obtained by photographing the flight attitudes of various aerial targets on the outdoor ground and consists of

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TABLE I

DATASET COMPOSITION AND RELATED PARAMETERS USED IN THE EXPERIMENT

dataset Type	Sequence number	Frame number	Type of interference	Maneuver type
Simulation	36	32707	infrared decoys	left maneuver, right maneuver, dive, climb
Measurement	9	7858	background clutter, infrared detector noise	roller and scale variation, maneuver variation

nine image sequences including complex cloud background interference, detector clutter interference, target maneuvering turns, attitude rotation, and other challenging situations. The keypoint coordinates of 32 346 aerial targets were annotated for a total of 7858 targets.

According to the test situation, 36 sets of simulated data and nine sets of measured data were selected for training and testing, as presented in Table I:

During network training, the loss function weight parameter is set to $\lambda_1 = 0.3$, $\lambda_2 = 0.5$, $\lambda_3 = 1$, and epoch $= 140$. Each batch of input 32 images trains the entire framework. The initial learning rate is set to 1e-3 and decays in a cosine fashion. The parameters are back forward propagation with the Adam optimizer. The proposed model implements network development in PyTorch [50], and the experiments are performed on an Nvidia GeForce GTX 3090 with 24 GB of memory.

We used the state-of-the-art pose estimation methods to evaluate the performance of the proposed model, and keypoint detection methods were selected based on two detection frameworks to evaluate the performance of the proposed model, including the bottom-up frameworks: high-resolution net (HRNet) (W32, W48) [51], HigherHRNet (W32, W48) [52], associative embedding (AE) [53], OpenPose [21]; top-down frameworks: ResNet [54], ResNext [55], Mask region CNN (RCNN) [23], AlphaPose [41], SimplePose [42], and Graph-pulse coupled neural network (PCNN) [56]. For fair comparisons, we directly run the public code of Mask RCNN and SimplePose [41] and retrain AlphaPose+. We use renet-101 as the backbone and AlphaPose with Yolo V3 as the detector for the pose estimation stage.

B. Evaluation Matric

During training, in the labeling process for each target, the form of the standard keypoint is defined as $[x_1, y_1, v_1, \ldots, x_K, y_K, v_K]$, where x and y are the coordinates of the keypoint, and v is the visibility flag. Here, $v = 0$ denotes unlabeled; $v = 1$ denotes labeled but not visible (such as occlusion); $v = 2$ indicates labeled, and the image is visible. Each standard object has a scale s, which is defined as the segmentation area of the object.

When testing, for each test result of the target, the keypoint detector should output the keypoint location and target confidence. The output format of the predicted keypoints should be the same as the standard keypoint form $[x_1, y_1, v_1, \ldots, x_K, y_K, v_K]$.

Based on point annotations and test results, evaluation metrics are proposed in terms of similarity. When evaluating the dataset comparison algorithm, this study only focused on the object keypoint similarity (OKS) of the target: $\text{OKS} = ((\sum_i \exp(-d_i^2/2s^2t_i^2) \cdot \delta(v_i > 0))/(\sum_i \delta(v_i > 0)))$.

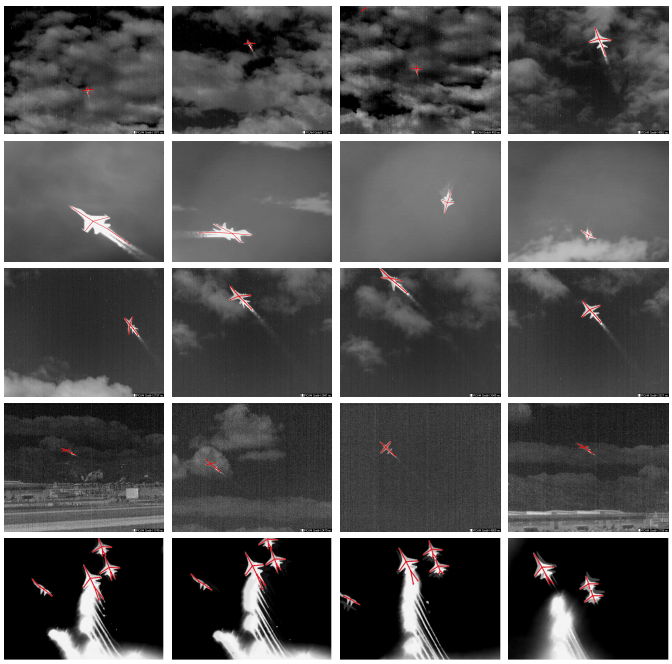


Fig. 8. Recognition results of measured infrared datasets. (First Row) Scale variation. (Second Row) Variation in type of maneuver. (Third Row) Roller rotation. (Fourth Row) Interference of cloud background, cartographic background, clutter of detector. (Fifth Row) Interference of infrared decoys.

where d_i is the Euclidean distance between the ground-truth point and the predicted point, and v_i is the visibility flag of the standard point. To calculate OKS, pass d_i through a nonstandard Gaussian function with standard deviation sk_i, where s is the scale of the target and t_i is a constant for each keypoint (to control the decay). A similarity ranging between 0 and 1 is generated for each keypoint. These similarities are averaged for all the keypoints of $v_i > 0$. Unmarked ($v_i = 0$) predicted keypoints do not affect the OKS indicator. A perfect prediction will make $\text{OKS} = 1$, whereas if all the keypoints in the prediction deviate by more than the standard deviation, then sk_i will make $\text{OKS} = 0$.

Based on the similarity of keypoints, the accuracy of aerial target interest point prediction is calculated and follows the standard evaluation index of Microsoft Common Objects in Context (MSCOCO) [57], which has been widely used in the existing studies. Based on the aforementioned keypoint similarity evaluation results, average precision (AP) and average recall were used to evaluate the algorithm. The IoU value selected by AP is 0.5:0.95, 0.5, and 0.75. To calculate the pose that satisfies the requirements of the algorithm training procedure, two rules were formulated to select the scheme. The target pose prediction should contain more than three visible points, and the OKS value should be greater than 0.25 to ensure quality. Finally, the visualization results of pose estimation and object recognition are given, as displayed in Figs. 8 and 9.

C. Qualitative Analysis

As displayed in Figs. 8 and 9, the entry angle represents the angle between the initial launch direction of the missile and

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TABLE II
COMPARISON OF ANTI-INTERFERENCE RECOGNITION RESULTS BASED ON KEY POINT ESTIMATION

Model	Backbone	mAP	AP^{50}	AP^{75}	AP^{M}	AP^{L}	AR
Top-down model							
AE[53]	—	0.615	0.808	0.663	0.556	0.696	0.672
OpenPose[21]	VGG19	0.618	0.769	0.615	0.581	0.682	0.675
HRNet(W32)[51]	HR32	0.697	0.838	0.681	0.655	0.758	0.740
HRNet(W48)[51]	HR48	0.704	0.851	0.692	0.661	0.757	0.746
HigherHRNet(W32)[52]	HR32	0.709	0.841	0.706	0.652	0.761	0.749
HigherHRNet(W48)[52]	HR48	0.715	0.847	0.713	0.669	0.765	0.755
Down-top model							
ResNet[54]	Res101	0.618	0.819	0.675	0.571	0.682	0.665
ResNext[55]	Res101	0.649	0.855	0.713	0.623	0.700	0.697
Mask-RCNN[23]	R50-FPN	0.631	0.833	0.687	0.578	0.714	—
AlphaPose+[41]	Res101	0.698	0.862	0.741	0.695	0.802	0.732
SimplePose[42]	Res101	0.631	0.865	0.712	0.612	0.685	0.698
Graph-PCNN[56]	HR48	0.707	0.854	0.716	0.610	0.687	0.703
MSGSM(Ours)	HR48	**0.738**	**0.874**	**0.752**	**0.704**	**0.817**	**0.723**

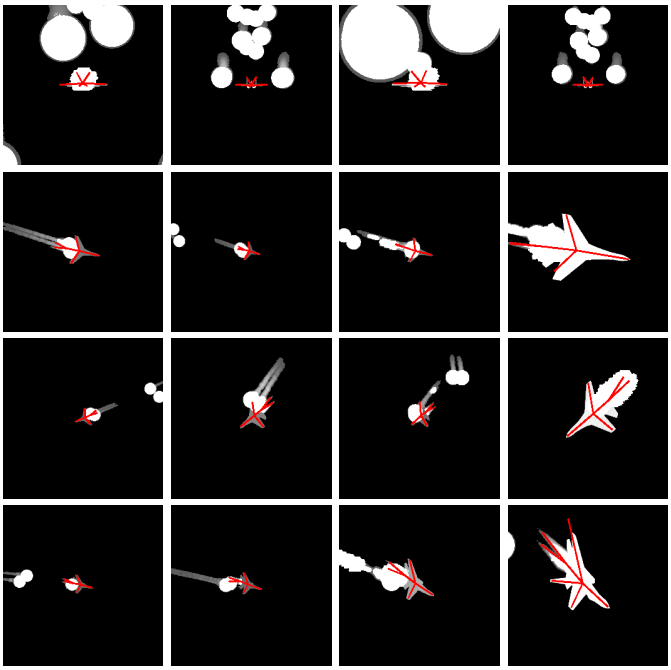


Fig. 9. Anti-interference recognition results in the infrared simulation datasets. The target maneuver type corresponding to each row of image sequence is the rear view of the aircraft, right maneuver, left maneuver, and dive maneuver, respectively, and its entry angle is 0°, 60°, 30°, and 90°, respectively.

the central axis of the air target, which indicates the initial direction of the missile attacking the target. The proposed MSGSM can correct faulty connections between nodes and estimate occluded nodes while maintaining high performance for visible nodes. The following conclusions can be drawn from the results: 1) for samples with object scale variation, roller rotation, and variation in type of maneuver, the proposed model provides excellent recognition results. Even errors with large displacements, large-scale changes, and fast drum rotations can be corrected with MSGSM; 2) furthermore, the second case includes the difficulty of ground object background and cloud background interference. However, the proposed method can adjust the joints to the correct position; and 3) the third aspect includes the case of dropping artificial decoy bombs, which reveals a proof that the proposed MSGSM module produces a natural posture. Even if the target is severely blocked by interference, the target node position can still be inferred accurately. The physical constraints of the target critical area show that under occlusion, the proposed method can obtain the correct connections between joints.

D. Quantitative Analysis

The quantitative results are presented in Table II. The results revealed that the proposed MSGSM achieved the best mAP compared with all the baselines with a considerable gap. MSGSM achieved significant gains with an astonishing 4.0 mAP@0.5:0.95 improvement compared with AlphaPose+. The MSGSM achieved a significant gain of 1.1 AP^{75}, which proved that MSGSM exhibits strong inference ability. These results validate the remarkable effectiveness of the proposed MSGSM model in anti-interference recognition.

The proposed MSGSM was compared with the best performing methods in its class. As presented in Table II, the method with HR48 backbone achieved the best mAP (73.8) and considerably improved HR48 (1.3) under the same input size (71.5). Furthermore, the proposed method outperformed other competitors with the same backbone and input size settings, such as HigherHRNet (71.5), Graph-PCNN (70.7), and HRNet (70.4). The highlight of the proposed method is highlighted.

E. Ablation Study

To analyze the proposed model in detail, a comprehensive ablation study was performed to evaluate the capabilities of each module. As presented in Table III, we provided the baseline to investigate the effect of each component. We denoted the keypoint-guided spatial correlation module as SCG, the skeleton model-guided GCN-based HRL module as HRL, and the HTL module as HTL. To investigate the effectiveness of the proposed framework and the contribution of each module, the experiments were structured by adding

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TABLE III
ABLATION STUDY OF THE ANTI-INTERFERENCE RECOGNITION MODEL

Models	Baseline	SCG	HRL	HTL	Stage1 mAP	Stage2 mAP
MSGSM_L11	✓	×	×	×	65.1	-
MSGSM_L12	✓	✓	×	×	68.9	-
MSGSM_L21	✓	✓	✓	×	70.3	71.4
MSGSM_L22	✓	✓	✓	✓	71.6	74.1

one module at a time. Next, an anti-occlusion target pose estimation performance test was established on the dataset. The method demonstrates the effectiveness of the GCN-based multisemantic anti-occlusion pose estimation models because of mAP of other methods.

1) *MSGSM_L11:* This method follows the heatmap-based keypoint detection framework and uses general keypoint heatmap detection (HR48) for rough estimation;
2) *MSGSM_L12:* Based on the keypoint detection method, the SCG module is added to associate the overall skeleton features of the target instead of scattered nodes. In the SCG module, GCN is used to extract target skeleton features;
3) *MSGSM_L21:* In this method, the proposed high-order relation learning module HRL is used after the spatial association module to strengthen the high-order relationship information (graph edge features);
4) *MSGSM_L22:* This is the complete model, which incorporates the proposed HTL on the basis of the third stage. This stage is used for inference of invisible keypoints and refines the results of target pose estimation.

The effectiveness of the proposed two-stage anti-interference recognition framework was evaluated. As presented in Table III, compared with the interest point detection of stage 1, the mAP index of the stage 2 is improved by 1.1 and 2.5, respectively, which indicates that the regression-based two-stage anti-interference recognition framework is compared. Therefore, the regression of the location of interest points in direct heatmaps is effective. After the overall framework is trained, stage 1 is extracted for testing. The training of the proposed model improves the performance of heatmap estimation of interest points in stage 1, which is 3.8, 5.2, and 6.5 mAP gain compared with the training of the first stage network alone.

Furthermore, we studied each design of MSGSM. The performance of MSGSM_L12, L21, and L22 reveals that the progressive, coarse-to-fine feature learning strategy is effective. Furthermore, the proposed cascaded multisemantic modules were analyzed. In Table III, the mAP values of the dataset decreased considerably, which indicated that the HRL module plays an integral role in the framework. We removed the HTL module and presented the results in Table III. The results proved the effectiveness of the reasoning part of the HTL high-order semantic learning module. Thus, image guidance is the most critical in the framework. The average gain of the HTL module on the datasets is 2.7 mAP, which reveals the need to adapt the image features to the inference module. Overall, these ablation studies overwhelmingly verified that each component is effective and that these cues inform invisible joint inference.

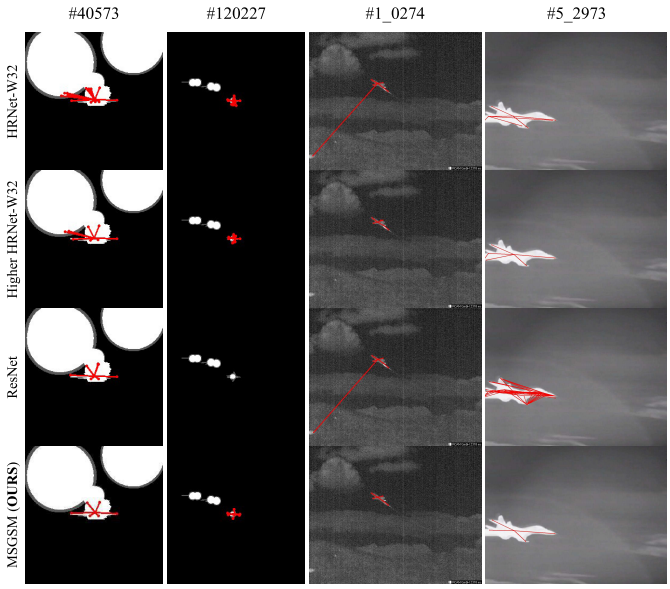


Fig. 10. Visualization of the target keypoint estimation results of our MSGSM model, HigherHRNet, HRNet, and ResNet for the challenging case of infrared datasets. Each column from left to right represents the scene of interference occlusion, distant target, cloud background, and close target, respectively.

F. Comparison of Visual Results

To evaluate the capability of our model to adapt to sophisticated scenarios, we illustrate a side-by-side comparison of our MSGSM model with the state-of-the-art approach in Fig. 10. Each column describes different scene comparison results, including fast motion, nearby people, occlusion, and video defocus, while each row shows the detection results of different methods. We compare the MSGSM model with three state-of-the-art methods, namely, HigherHRNet-W32 [52], HRNet-W32 [51], and ResNet [54]. We observe that our approach yields more robust detection for this challenging situation. No matter in the measured or simulation datasets, the three compared algorithms all have different degrees of false detection, missed detection, and repeated detection. In particular, in the first column of images, when the target is severely occluded, both HigherHRNet and HRNet have different degrees of misdetection, and our model effectively estimates the key position of the target. Our proposed MSGSM model can encode the spatial information of nodes better through the principle design of HRL and HTL modules and refine the detection of target pose through GCN. Our method has reached a new advanced level both quantitatively and qualitatively.

VI. CONCLUSION

We proposed a novel MSGSM model to address the occlusion problem in infrared target anti-interference recognition. Three well-designed components, an image keypoint-guided GCN spatial correlation module, and a cascaded high-order semantic information inference module were obtained for accurate target keypoint estimation and anti-occlusion inference. Comprehensive experiments were conducted on our datasets and ablation studies to detail its effectiveness and provide various insights. The experiments revealed that the

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heatmap and GCN-based modules work synergistically to achieve considerable improvements in all aspects.

Our proposed infrared aerial target anti-interference recognition model still has some shortcomings and deficiencies, mainly including two points: on one hand, our proposed MSGSM model infers and predicts the occluded part of the target by learning the skeleton structure of the target, without considering serious occlusion or even full occlusion. On the other hand, the MSGSM model only tries to understand the spatial positional relationship of key points in a single frame of pictures, lacks learning the node change relationship and characteristics between two or more frames, and cannot make full use of the labeled data. These are the directions we need to improve and strengthen in the future. We hope that our work is a boost research on occlusion problems in infrared aerial target anti-interference recognition.

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